

Achraf **EL ALLALI**

FINAL-YEAR INTERNSHIP | SOFTWARE ENGINEERING

FULL STACK · GENERATIVE AI · DEVOPS

Contact

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Skills

Languages

C# · Python · Java · C/C++ · PowerShell

Frontend

Svelte 5 · TypeScript · Angular · React · JavaScript · HTML/CSS · Tailwind

Backend

FastEndpoints · Spring Boot · .NET · Node.js · FastAPI · Flask · REST API

AI & Data

RAG · LLM · Vector DB · PostgreSQL · Oracle · SQLite · Embeddings · MySQL

DevOps & Cloud

Docker · Kubernetes · CI/CD · Linux · Scaleway · Azure Blob

Tools

Git · Blender · Unity · STM32Cube

Languages

French	Fluent
English	Fluent
Arabic	Native

Qualities

Curious
Resourceful
Making things work

Interests

Chess
Sports
Reading

Professional Experience

Software Developer — *France*

CTCPA — *Sept. 2026 – Feb. 2027*

- Continued development of the **IoT** platform HeatSight.
- Implemented observability features: application metrics, structured logging and error reporting in **Sentry**.
- Programmed sensors in **C** with **STM32Cube**.
- Improved **CI/CD** pipelines and optimized resources on **Scaleway**.
- Contributed to algorithmic modules, cybersecurity reviews and infrastructure work.
- IT support: device management with **Intune**, identity management with **Microsoft Entra ID** (Azure AD) and **WatchGuard** firewall administration.
- **Keywords:** C#, .NET, Svelte 5, C, STM32Cube, Sentry, CI/CD, Scaleway, S3, Docker, PostgreSQL, IoT, Networking, Intune, Entra ID, WatchGuard

AI Banking Hackathon – 3rd place

Sopra Steria x Crédit Agricole — *Oct. 2026*

- Won 3rd place as a team with Smith, a banking assistant.
- **Angular** frontend and **FastAPI** backend in **Python**.
- Deterministic engine (Safe to Spend), custom **LangChain LLM** and **RAG** with **ChromaDB** and **Hugging Face**.
- Simulated data in **SQLite** and **JSON**.
- **Keywords:** Angular, FastAPI, Python, LangChain, LLM, RAG, ChromaDB, Hugging Face, Embeddings, SQLite, JSON, Generative AI

Full-Stack Developer Intern — *Avignon, France*

CTCPA — *Mar. 2026 – Aug. 2026*

- Optimized the **CI/CD** chain: multi-environment pipelines, quality gates and automated deployment.
- Containerized backend/frontend services with **Docker** for development and staging environments.
- Designed and developed **IoT** modules: semi-log scale, booking, sensor calibration and adjustment.
- Built **FastEndpoints** APIs using a Vertical Slice architecture.
- Developed the frontend (**Svelte 5**, **Tailwind**) and a **.NET Core** REST backend following an offline-first **PWA** architecture.
- **Keywords:** Scaleway, S3, Docker, C#, .NET, FastEndpoints, Svelte 5, Tailwind, PostgreSQL, SQLite, WebAssembly, CI/CD, IoT

Projects

OCULA: Generative AI Video Analysis Web App

Personal project – Team of 3 — *Feb. 2026*

- Designed and developed a web application to upload videos and query their content through a chat-bot.
- Built a **RAG**-based AI pipeline: audio transcription, text chunking, **embedding** generation and semantic search.
- Containerized the services (**Docker**), deployed the pipeline on **Azure Blob Storage** and integrated **CI/CD**.
- **Keywords:** Angular, Spring Boot, Flask, Python, Java, REST API, Docker, PostgreSQL, CI/CD, Git, Azure Blob Storage

Education

Engineering Degree in Computer Science — *Clermont-Ferrand, France*

ISIMA — *2024 – 2027*

Specialization: Software Engineering and Computer Systems

Bachelor's Degree in Computer Science — *Clermont-Ferrand, France*

Université Clermont Auvergne — *2022 – 2024*

Specialization: Mathematics and Computer Science